ZOO403 ANIMAL BEHAVIOR

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Parental care:

Parental care can be defined as any non-genetic contribution by a parent that increases the fitness of offspring, and can occur before or after laying or birth.

Parental care in mammals:

In mammals, lactating female nourishes the young and is involved in most of the caring behaviors. Males usually play no role in raising the young. In some mammalian species, such as lions, males protect the females and young. In such species a male or small group of males takes care of many females at once in a harem.

How polygynous occur if there is no parental care is required:

Males in mammals and birds do not stay with their partner if young ones can feed and care for themselves after hatching. Males of these species maximize their reproductive success by seeking other mates. Polygyny is more common in such animals.

How verve monkey show social behavior: long

Vervet monkeys produce a complex set of alarm calls. They give distinct alarm calls for predators like leopards, eagles or snakes. Alarm call for leopard is a loud barking sound. For an eagle a short double-syllabled cough. For a snake, a “chutter”. Upon hearing an alarm call, other vervets behave in an appropriate way: Run up a tree on hearing alarm call for leopard, Look up on hearing alarm for an eagle, Look down on hearing alarm for a snake. Infant vervets give and understand alarm calls, but in a relatively undiscriminating way. They give “eagle alarm” on seeing any bird. With age, they fine-tune their calls by learning from other members of the group.

Diffirent Types of territory:

Mating Territory: places of mating for the partner.

Breeding Territory: The breeding territory is relatively small. It usually contains only a nesting or mating site.

Feeding Territory: It tends to be larger than the breeding territory because it must contain sufficient food to support the owner of the territory and any mate or offspring that may also be residing there.

Multi- purpose Territory: This type of territory is generally the largest as it includes aspects of both breeding and feeding territories.

What is the Difference between polygyny and polyandrous system:

Polygyny system	Polyandrous system
Polygynous systems involve an exclusive relationship of one male with two or more females.	In polyandrous systems, one female has an exclusive relationship with two or more males.

Polygynous species are generally dimorphic, with males being showier and often larger than females.	Polyandrous species are also dimorphic with females generally more ornamented and larger than males.

Parental care in birds:

Birds show maximum parental care. Most newly hatched birds cannot care for themselves. They require a large, continuous food supply. Catering the needs of young is difficult for a single parent. To enhance chances of viability of offspring, most birds form monogamous pairs. Male stays with its mate and helps her till offspring are able to fulfill their own needs. Lactating female nourishes the young. She provides most of the required care.

What are the Cognitive maps, how the birds show cognitive maps behavior:

Cognitive Maps: "A representation in the nervous system of the spatial relationships between objects in an animal's surroundings". It involves relating landmark positions to one another. Cognitive maps follow abstract geometric rule—identifying halfway point between landmarks. Reduce the amount of detail required to remember an object's location.

Cognitive Map—Example: Clark's nutcracker birds. Store thousands of pine seeds in fall at thousands of hiding places. Caches—distributed over a large area (35 km^2). During winter, they relocate their caches.

Types of cues used in migration and navigation:

Navigational abilities rely on several sensory cues (signals). Guide in orientation and path finding. Some types of cues are not perceivable by human beings.

Migratory and Navigational Cues: 1= Sun compass 2= Celestial cues 3= Earth's magnetic field 4= Visual cues 5= Auditory cues 6= Olfactory cues.

Advantages of territorial behavior:

1= Territorial behavior forms an important part of social behavior. 2= Territory provides a base which the male can advertise himself to female, and attracted the female. 3= Provides a protected place for courtship, mating and parental care to young. 4= Provided safety where young can live freely and without any interference.

Define cognition:

"The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments". It is the Most complex form of learning, also Characteristically exhibited by primates.

Define Foraging behavior:

The behavior that includes all activities of an animal related to search, recognize and capture food items is called foraging behavior.

Define Altruism:

Altruism is a selfless behavior in which an individual endangers his own survival or sacrifices some of its own reproductive potential to benefit another individual.

Social organization:

A social organization is a large group of members who intercommunicate extensively, exhibit division of labour, bear some relatively permanent social relationship to one another, Such organization exists in both invertebrates and vertebrates. arthropods such as ants, bees and termites are the best examples of the social organization.

In list the animals that show altruistic behavior:

Social insects such as ants, wasps, bees and termites and squirrel, stinging bees, Vampire bats, Emperor penguins, Vervet monkeys, Cleaner fish, Primates, Wolves, dolphins.

Intersexual selection:

It Involves mate choice. Members of one sex choose mates on the basis of particular characteristics of other sex, such as appearance, color or courtship songs.

Inclusive Fitness:

Inclusive fitness is the measure of fitness of a group of members of a species—which may not be close relatives. An organism's inclusive fitness is determined by its personal fitness, plus its weighted effects on the fitness of every other individual in population.

Competition:

Competition is the struggle among organisms to survive in a habitat with limited resources (species searching for the same kinds of food). Competition happens when organisms in an ecosystem have similar needs. Organisms with different needs can live and survive together in a habitat with little competition. Competition increases when the amount of available resources is low. Members of the same species compete for available resources. All organisms compete for resources (not just animals).

Types of behaviors in animals:

There are 2 basic types of behavior 1= Innate behaviors or Inherent behavior 2= Learned behaviors or acquired behavior. Nature/innate: Instinct or genes determine behaviour. Nurture/learned: Experience and learning influence behaviour.

Intraspecific or interspecific:

If you break the two terms down, "intraspecific" just means within a species, while "interspecific" means between them. Consequently, interspecific competition is all about competition between two or more species, while intraspecific competition involves different individuals of the same species. Interspecific Communication: Communication among different species, Prey to predator e.g., warning coloration in wasps, Predator to prey e.g., Angler fish, Human/animal communication during domestication.

Reflex action:

Reflex action is a sudden and involuntary response to stimuli.

Dominance her aches in wolf pack:

Dominance is highly organized in these species. These animals hunt in groups and are keenly aware of their own status in relation to the other individuals present within the group.

Examples: Dominant wolves are referred to as alphas. There are commonly both alpha males and females. The alpha status assures a wolf the highest priority in getting every resource e.g. food. These factors determine the survival of a particular individual and their ability to pass on their genes.

Altruism in animals:

Evolutionary biologists determined that an animal's behaviors are altruistic when they benefit other individuals, even to the potential detriment of themselves. Altruistic behaviour is common throughout the animal kingdom, particularly in species with complex social structures. For example, vampire bats regularly regurgitate blood and donate it to other members of their group who have failed to feed that night, ensuring they do not starve.

Cross-fostering:

Cross-fostering is a technique used in animal husbandry, animal science, genetic and nature versus nurture studies, and conservation, whereby offspring are removed from their biological parents at birth and raised by surrogates, typically of a different species, hence 'cross.'

Two examples of animal cognition:

1= Clark's nutcracker birds. Store thousands of pine seeds in fall at thousands of hiding places. Caches—distributed over a large area (35 km^2). During winter, they relocate their caches. 2= Ants use dead reckoning to locate their nests after convoluted foraging trips.

Dominance hierarchy:

a dominance hierarchy (formerly and colloquially called a pecking order) is a type of social hierarchy that arises when members of animal social groups interact, creating a ranking system. A dominant higher-ranking individual is sometimes called an alpha, and the submissive lower-ranking individual a beta.Dominance hierarchies are best known in social mammals, such as baboons and wolves, and in birds, notably chickens.

Kin selection:

It is an altruistic evolutionary strategy that favors the reproductive success of an organism's close relatives, even at a cost of organism's own survival and reproduction. William Hamilton proposed the idea of kin (relatives) selection to explain how selection acting on related animals can affect the fitness of an individual. Through kin selection, a gene carried by a particular individual passes to the next generation through a related individual.

Innate behavior:

Innate behavior is also called instinctive behavior. Instinct is the ability of an animal to perform a particular behavior in response to a given stimulus the first time the animal is exposed to the stimulus. In other words, an instinctive behavior does not have to be learned or practiced. For example birds nesting behavior.

Examples of Social Animals:

1= Ants and termites 2= Bees and wasps 3= Many birds 4= Wildebeests 5= Wolves and Lions 6= Monkeys and gorillas 7= Humans

Difference between intersexual selection and intrasexual selection:

intersexual selection	intrasexual selection
Intersexual selection involves mate choice. In this selection, members of one sex choose mates on the basis of particular characteristics of other sex, such as appearance, color or courtship songs.	Intrasexual selection involves competition between members of the same sex for access to members of the opposite sex.
Intersexual selection has led to sexual dimorphism in secondary sexual characteristics, such as the ornate plumage of birds, or the antlers of deer, or the manes of lions.	Intrasexual selection usually occurs between males and may take the form of male-to-male combat or fight.
Such selective characters correlate in general with the mate's health and vitality.	Such species or individuals have developed better weapons for competition e.g. horns, antlers etc.

Gains and Costs of Foraging:

There are energy gains and costs in finding and consuming food: 1= Energy gain from food 2= Energy expenditure in foraging 3= Risk of predation while foraging.

Characteristics of social behavior:

Social behaviour ranges from simple attraction between individuals to life in complex societies characterized by division of labour, cooperation, altruism, and a great many individuals aiding the reproduction of a relative few. The most widely recognized forms of social behaviour, however, involve interaction within aggregations or groups of individuals. Social behaviours, their adaptive value, and their underlying mechanisms are of primary interest to scientists in the fields of animal behaviour, behavioral ecology, evolutionary psychology, and biological anthropology.

Plasticity of human behavior:

Plasticity can be defined as the brain's capacity to achieve lasting structural changes in response to environmental demands that are not fully met by the organism's current functional capacity.

Just like environment intervenes in the phenotypic expression of a genotype for physical traits, it is more significantly involved in the expression of behavioral traits. As human have greater capacity for learning, human behavior is probably more plastic than that of any other animal.

Salient features of territoriality:

1. Definite home and home range 2. Adaptive Features of Territoriality 3. Food and security 4. Marking and Defending Territories.

Theories of male behavior defending best of neighbors:

1=Males only defend other nests which contain their extrapair offspring. 2= Another is the titfor-tat strategy of reciprocal altruism. 3= A third theory is, males help only other closely related males.

Difference between Specialist and generalists:

Specialist	generalists
species are animals that require very unique resources. Often, these species have a very limited diet, or need a specific habitat condition to survive.	Species that can live in many different types of environments, and have a varied diet are considered generalists.
Specialist species evolved to fit a very specific niche. • Environmental disruptions like effects from climate change or habitat loss have a strong effect on specialists.	• Generalists can eat a variety of foods • Thrive in a range of habitats. • They have broad niches. • They can thrive in rapidly changing environmental conditions.

Reproductive behavior in Ring dove:

Male and female ring doves coordinate their participation in: 1= courtship 2= nest building 3= Incubation 4= care of young

Reciprocal altruism:

Reciprocal altruism (according to Trivers) is altruism that occurs between unrelated individuals when there will be repayment (or at least the promise of repayment) of the altruistic act in the future.